Gyuwon Na

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EDUCATION

DigiPen Institute of Technology | Redmond, WA — **Aug 2026 – Expected May 2028**
B.S. Computer Science in Real-Time Interactive Simulation | GPA: 4.0 / 4.0

Keimyung University | Daegu, South Korea — **Mar 2023 - Jun 2026**
KMU-DigiPen Dual-Degree Program | GPA: 4.38 / 4.5
Coursework: Computer Graphics, Data Structures, Computer Networks, Digital Signal Processing, Circuit Analysis

TECHNICAL SKILLS

Languages: C++, Python, C, GLSL
Tools & Platforms: Linux, CMake, Git/GitHub, ROS 2 / ROS 1, Docker, MATLAB
Libraries & Engines: OpenGL, WebGL, OpenCV, PyTorch, CUDA, Unity

RESEARCH EXPERIENCE

ISIP Lab | Undergraduate Researcher | Daegu, South Korea — **Aug 2025 - Present**

- Designed a **1.43M-parameter, five-stage geometric alignment model** using Clifford-inspired geometric representations, transformer-based estimation, and residual/energy-based refinement for severe in-plane rotations.
- Reduced mean rotation error on the **100 hardest recovery cases from 40.49° to 17.01° (58%)**; evaluated across COCO, HPatches, and MegaDepth with fixed-angle stress tests, baselines, ablations, and failure-case analysis.
- Developed an **ROI-constrained ORB/HNSW feature-matching pipeline** with MAGSAC++ verification, reducing candidate comparisons by **96.12% while retaining a 0.99 matching success rate.**
- Developed a lightweight lane-detection pipeline using **morphological closing and adaptive Otsu thresholding**, reducing partial-shadow noise penalty from **73.7 px to 9.1 px** and noise pixels by approximately **30%**.
- First author of an **Image and Vision Computing** manuscript under review and a **KMMS Fall Conference** poster on adaptive lane detection.

PROJECTS

SEA:ME Hackathon - Vision-Only Driving System | Lead Autonomous Systems Engineer — **Jul 2026**

- Led integration of a **single-camera ROS 2 autonomy stack** using front-camera perception for lane tracking, traffic-light/sign recognition, mission-state logic, Pure Pursuit control, and actuator output.
- Increased perception throughput from **6 FPS to a stable 20 FPS (3.3x)** through shared-memory process isolation, mission-specific ROIs, fresh-frame queues, and ROS 2 execution tuning.
- Completed the course in **56 seconds**, finishing only **0.03 seconds** behind 3rd place.

LEGO Finder - Prompt-Guided Visual Search | Personal Project — **Apr 2026 - Jun 2026**

- Built a **real-time prompt-guided visual search pipeline** that localizes candidate regions with ROI masking, retrieves local-feature matches from an offline **HNSW** index.
- Addressed **partial occlusion and visually similar distractors** by combining color/shape cues with **RANSAC-based geometric verificatio**n to reject ambiguous feature matches.

KAIST Mobility Challenge - V2X Cooperative Driving | Systems Engineer — **Oct 2025 - Feb 2026**

- Built V2X-based cooperative decision and control logic for **four autonomous vehicles** negotiating roundabouts and four-way intersections, using shared pose and velocity states for real-time collision avoidance and stop-or-go decisions.
- Implemented **MPC-inspired Pure Pursuit tracking**, sim-to-real sigmoid braking compensation, and **YAML-configured per-vehicle tuning** to account for vehicle-specific dynamics and streamline field deployment.

MANZO - Custom C++ / OpenGL Game Engine | Technical Lead — **Jun 2024 - Jun 2025**

- Led a **year-long engine and game project**, implementing rendering system include vector-font rendering, underwater visibility shaders, reusable UI systems, positional audio, and debugging the rendering pipeline with **RenderDoc**.
- Engineered autonomous fish behaviors with pathfinding, terrain avoidance, and flocking, using quadtree-based spatial partitioning to reduce unnecessary per-frame spatial queries.

Stand-By - Custom C++ Game Engine | Technical Lead — **Mar 2024 - Jun 2024**

- Led the software architecture of a **C++ / Raylib cafe-management game**, extending a lightweight engine with reusable state, object, collision, and scene-management systems.
- Structured customer lifecycles and drink-making interactions as explicit state-driven systems with per-object timers and context-specific interaction paths, improving behavior consistency and debuggability across repeated game cycles.

AWARDS

SEA:ME Hackathon | Silver Award, 4th Overall — **Jul 2026**
KAIST Mobility Challenge | Excellence Award, 3rd Overall — **Feb 2026**
2025 HL Future Mobility Award | Encouragement Award — **Sep 2025**
2025 Micro Degree Expo | Outstanding Award, 2nd Place — **Jun 2025**
Artificial Intelligence Software Competition – C Programming Division | Excellence Award, 3rd Place — **Aug 2024**